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Equity Valuation: Concepts and Basic Tools

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DIVIDEND DISCOUNT MODEL (DDM) AND FREE-CASH-FLOW-TO-EQUITY MODEL (FCFE)

Learning Outcomes

- explain the rationale for using present value models to value equity and describe the dividend discount and free-cash-flow-to-equity models
- explain advantages and disadvantages of each category of valuation model

If the issuing company is assumed to be a going concern, the intrinsic value of a share is the present value of expected future dividends. If a constant required rate of return is also assumed, then the DDM expression for the intrinsic value of a share is Equation 1:

$$V_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1+r)^t} \qquad 1$$

where

V_0 = value of a share of stock today, at $t = 0$

D_t = expected dividend in year t, assumed to be paid at the end of the year

r = required rate of return on the stock

At the shareholder level, cash received from a common stock investment includes any dividends received and the proceeds when shares are sold. If an investor intends to buy and hold a share for one year, the value of the share today is the present value of two cash flows—namely, the expected dividend *plus* the expected selling price in one year:

$$V_0 = \frac{D_1 + P_1}{(1+r)^1} = \frac{D_1}{(1+r)^1} + \frac{P_1}{(1+r)^1} \qquad 2$$

where P_1 = the expected price per share at $t = 1$.

To estimate the expected selling price, P_1, the analyst could estimate the price another investor with a one-year holding period would pay for the share in one year. If V_0 is based on D_1 and P_1, it follows that P_1 could be estimated from D_2 and P_2:

$$P_1 = \frac{D_2 + P_2}{(1+r)}$$

Substituting the right side of this equation for P_1 in Equation 2 results in V_0 estimated as

$$V_0 = \frac{D_1}{(1+r)} + \frac{D_2 + P_2}{(1+r)^2} = \frac{D_1}{(1+r)} + \frac{D_2}{(1+r)^2} + \frac{P_2}{(1+r)^2}$$

Repeating this process, we find the value for n holding periods is the present value of the expected dividends for the n periods plus the present value of the expected price in n periods:

$$V_0 = \frac{D_1}{(1+r)^1} + \cdots + \frac{D_n}{(1+r)^n} + \frac{P_n}{(1+r)^n}$$

Using summation notation to represent the present value of the n expected dividends, we arrive at the general expression for an n-period holding period or investment horizon:

$$V_0 = \sum_{t=1}^{n} \frac{D_t}{(1+r)^t} + \frac{P_n}{(1+r)^n} \qquad 3$$

The expected value of a share at the end of the investment horizon—in effect, the expected selling price—is often referred to as the **terminal stock value** (or **terminal value**).

Estimating Share Value for a Three-Year Investment Horizon

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1. For the next three years, the annual dividends of a stock are expected to be €2.00, €2.10, and €2.20. The stock price is expected to be €20.00 at the end of three years. If the required rate of return on the shares is 10 percent, what is the estimated value of a share?

Hide Solution

Solution:

The present values of the expected future cash flows can be written as follows:

$$V_0 = \frac{2.00}{(1.10)^1} + \frac{2.10}{(1.10)^2} + \frac{2.20}{(1.10)^3} + \frac{20.00}{(1.10)^3}$$

Calculating and summing these present values gives an estimated share value of V_0 = 1.818 + 1.736 + 1.653 + 15.026 = €20.23.

The three dividends have a total present value of €5.207, and the terminal stock value has a present value of €15.026, for a total estimated value of €20.23.

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Extending the holding period into the indefinite future, we can say that a stock's estimated value is the present value of all expected future dividends as shown in Equation 1.

Consideration of an indefinite future is valid because businesses established as corporations are generally set up to operate indefinitely. This general form of the DDM applies even in the case in which the investor has a finite investment horizon. For that investor, stock value today depends *directly* on the dividends the investor expects to receive before the stock is sold and depends *indirectly* on the expected dividends for periods subsequent to that sale, because those expected future dividends determine the expected selling price. Thus, the general expression given by Equation 1 holds irrespective of the investor's holding period.

In practice, many analysts prefer to use a free-cash-flow-to-equity (FCFE) valuation model. These analysts assume that dividend-paying *capacity* should be reflected in the cash flow estimates rather than *expected dividends*. FCFE is a measure of dividend-paying capacity. Analysts may also use FCFE valuation models for a non-dividend-paying stock. To use a DDM, the analyst needs to predict the timing and amount of the first dividend and all the dividends or dividend growth thereafter. Making these predictions for non-dividend-paying stock accurately is typically difficult, so in such cases, analysts often resort to FCFE models.

The calculation of FCFE starts with the calculation of cash flow from operations (CFO). CFO is simply defined as net income plus non-cash expenses minus investment in working capital. FCFE is a measure of cash flow generated in a period that is available for distribution to common shareholders. What does "available for distribution" mean? The entire CFO is *not* available for distribution; the portion of the CFO needed for fixed capital investment (FCInv) during the period to maintain the value of the company as a going concern is *not* viewed as available for distribution to common shareholders. Net amounts borrowed (borrowings minus repayments) are considered to be available for distribution to common shareholders. Thus, FCFE can be expressed as

$$\text{FCFE} = \text{CFO} - \text{FCInv} + \text{Net borrowing} \qquad 4$$

The information needed to calculate historical FCFE is available from a company's statement of cash flows and financial disclosures. Frequently, under the assumption that management is acting in the interest of maintaining the value of the company as a going concern, reported capital expenditure is taken to represent FCInv. Analysts must make

$$V_0 = \sum_{t=1}^{\infty} \frac{\text{FCFE}_t}{(1+r)^t}$$

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EXAMPLE 5

Present Value Models

1. An investor expects a share to pay dividends of $3.00 and $3.15 at the end of Years 1 and 2, respectively. At the end of the second year, the investor expects the shares to trade at $40.00. The required rate of return on the shares is 8 percent. If the investor's forecasts are accurate and the market price of the shares is currently $30, the *most likely* conclusion is that the shares are:

   A. overvalued.
   B. undervalued.
   C. fairly valued.

   Hide Solution

   Solution to 1:

   B is correct.

   $$V_0 = \frac{3.00}{(1.08)^1} + \frac{3.15}{(1.08)^2} + \frac{40.00}{(1.08)^2} = 39.77$$

   The value estimate of $39.77 exceeds the market price of $30, so the conclusion is that the shares are undervalued.

2. Two investors with different holding periods but the same expectations and required rate of return for a company are estimating the intrinsic value of a common share of the company. The investor with the shorter holding period will *most likely* estimate a:

   A. lower intrinsic value.
   B. higher intrinsic value.
   C. similar intrinsic value.

   Hide Solution

   Solution to 2:

   C is correct. The intrinsic value of a security is independent of the investor's holding period.

3. An equity valuation model that focuses on expected dividends rather than the capacity to pay dividends is the:

   A. dividend discount model.
   B. free cash flow to equity model.
   C. cash flow return on investment model.

   Hide Solution

   Solution to 3:

   A is correct. Dividend discount models focus on expected dividends.

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How is the required rate of return for use in present value models estimated? To estimate the required rate of return on a share, analysts frequently use the capital asset pricing model (CAPM):

$$\begin{aligned}\text{Required rate of return on share } i = {} & \text{Current expected risk} \\ & \text{–free rate of return} \\ & + \text{Beta}_i \left[\text{Market (equity) risk premium}\right]\end{aligned}$$

market in excess of the risk-free return, where in practice, the "market" is often represented by a broad stock market index). However, even if analysts agree that the CAPM is an appropriate model, their inputs into the CAPM may differ. Thus, there is no uniquely correct answer to the question: What is the required rate of return?

Other common methods for estimating the required rate of return for the stock of a company include adding a risk premium that is based on economic judgments, rather than the CAPM, to an appropriate risk-free rate (usually a government bond) and adding a risk premium to the yield on the company's bonds. Good business and economic judgment is paramount in estimating the required rate of return. In many investment firms, required rates of return are determined by firm policy.

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Great read.

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Could somenone please explain why the intrinsic value is independent of the holding period. Wouldn't a longer holding period lead to more "dividend terms" in the Dividend Discount Model?

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